# THE FORGOTTEN VILLAGE

A VR Mystery RPG with a Physical Safety-Control Console

Course: Mechatronics and XR Lab (2026) | Project Type: VR × Mechatronics Integration
Team Members / Student IDs : 113511011 程婕茵 , 113511013 范芷祺

## 01 MOTIVATION AND OBJECTIVE

The Forgotten Village is a mystery role-playing game in which an amnesiac traveler investigates five village houses, solves engineering puzzles, and collects records of a vanished researcher. After unlocking the sixth, abandoned house, the player reconstructs an unsafe experiment and discovers that the missing researcher is the player.

The project extends this narrative into an interactive VR–physical system. A real control console becomes part of the fictional experiment: players adjust physical inputs to influence virtual instruments, while decisions in Unity drive a tabletop actuator. The goal is to demonstrate coherent bidirectional integration rather than isolated hardware effects.

## 02 EXEPECTED RESULTS

The expected outcome is a playable VR mystery RPG featuring six explorable houses, five engineering puzzles, and a complete investigation storyline. A functional Arduino-based control console will enable bidirectional interaction between physical hardware and Unity, demonstrating integrated sensor input, actuator control, and immersive problem-solving.

## 03 GAME CONCEPT AND ENGINEERING CONTENT

The five clue houses may be explored in any order; collecting all five clues unlocks the final investigation. Each puzzle contributes evidence about the same Memory Interface Controller experiment:

- Elder’s House — Voltage-divider loading effect; recover damaged circuit notes.
- Alchemist’s House — RC low-pass filtering; study the useful and unwanted signal bands.
- Carpenter’s House — MOSFET common-source amplifier gain; inspect the analog front end.
- Technician’s House — C++ object slicing; diagnose a safety controller that incorrectly outputs RUN.
- Clockmaker’s House — VLSI setup timing slack; identify a potential timing violation.
- Abandoned House — Combine the records, compare deployed software and runtime logs, and reveal the researcher’s identity.

## 04 HARDWARE AND VR SYSTEM CONFIGURATION

The baseline design uses a Meta Quest headset connected to a Windows PC through Quest Link. The Unity PC VR application communicates with an Arduino Uno over USB Serial. The Arduino reads a 10 kΩ potentiometer and push button and controls a small SG90 servo; an LED provides optional status feedback.

**Physical Console ↔ Arduino Uno ↔ USB Serial / Unity PC ↔ Meta Quest Link**

Physical → virtual: turning the potentiometer changes a simulated filter parameter in the Alchemist’s House; pressing the button triggers the controller diagnostic in the Technician’s House. Virtual → physical: a valid in-game safety command makes Arduino move a miniature safety gate using the servo in the final house.

## 05 INNOVATION AND PROMOTION POINTS

- Narrative-driven hardware: real electronics represent the missing researcher's experimental interface and reinforce the mystery.
- Bidirectional interaction: physical inputs affect VR diagnostics; a verified VR event produces an observable mechanical action.
- Cross-disciplinary learning: circuit analysis, signals, analog electronics, advanced OOP, and VLSI are linked through one evidence trail.
- Evidence-based resolution: deployed software and runtime logs substantiate the object-slicing failure, while negative timing slack remains a possible risk rather than a proven cause.

## 06 TEAM MEMBERS AND RESPONSIBILITIES

### 程婕茵 — Player Development & Gameplay Design

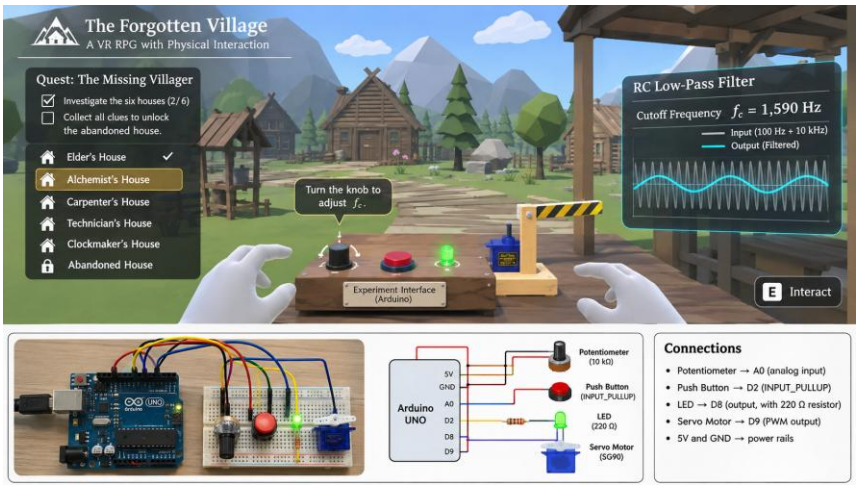
- Develop player movement, VR interactions, and object manipulation.
- Design and implement three houses
- Integrate Arduino sensor inputs into the corresponding gameplay mechanics.

### 范芷祺 — UI Systems & Hardware Integration

- Develop the dialogue system, inventory, clue journal, and quest UI.
- Design and implement the remaining three houses.
- Implement Arduino communication and integrate sensors.
- Assist with VR system integration and hardware testing.

## 07 DISCLOSURE OF AI ASSISTANCE

The project concept, system architecture, and hardware integration plan were developed by our team. ChatGPT was used solely to assist in organizing our ideas and expressing them clearly in written form. All hardware implementation, software development, system integration, and testing will be independently carried out by our team.




## 08 REFERENCE

Unity Documentation — https://docs.unity3d.com/

Meta Quest Developer Documentation — https://developers.meta.com/horizon/

Arduino Documentation — https://docs.arduino.cc/